Reproducing Figure 2 of "Chain-of-Thought Reasoning in the Wild Is Not Always Faithful", with two extensions

[Name 1], [Name 2], [Name 3] CS2881r mini-project, October 2026

Code, data and decision log: `https://github.com/kvrancic/cs2881r-iphr-reproduction`

Summary

Arcuschin et al. (ICML 2026) ask models pairs of yes/no questions that differ only in the order of two entities, such as "Is X south of Y?" and "Is Y south of X?". Exactly one is true. Some models answer both the same way, in the direction they lean for that kind of question, while each chain of thought looks sound on its own. The authors call this *implicit post-hoc rationalization* (IPHR). Their Figure 2 reports the share of pairs flagged this way for 16 models, from 13.5% (GPT-4o-mini) to 0.04% (Claude 3.7 Sonnet with thinking).

1. **Released data.** Our code gives the plotted counts exactly for the 8 models whose 10-answer counts were released, as well as all 64 values of the paper's Table 10 (rates over other cutoffs). We found three discrepancies in the paper: Figure 2 divides by 4,892 pairs where the data hold 4,834; the 100-answer retest uses a 0.4 cutoff where the text states 0.5 (at 0.5, 61% of flags survive; the paper reports 76.5%); and the text says the model ranking holds at every cutoff while its own Table 10 shows swaps.
2. **Fresh runs.** On 1,014 pairs with the paper's prompt and settings, each original model's lean gap is within 0.012 of the paper's. Flag counts agree for GPT-4o-mini and Llama 3.3 70B. DeepSeek V3 gets fewer, partly from our judge and partly because its leans sit at the cutoff. Pairs that showed extra asymmetry in the lean direction in the paper's run show it again in ours, clearly for GPT-4o-mini and weakly for the other three.
3. **By kind of question.** Geography leans NO and other questions lean YES. The model ranking nearly reverses between the two: GPT-4o-mini is flagged on 19.6% of geography pairs and 1.4% of the rest, QwQ on 0.2% and 13.6%.
4. **Extensions.** Printing both true values removes the flags, as expected. Printing one rounded value for both makes the models abstain (89–98% UNKNOWN) with no lean. Asked for each value alone, Llama 3.3 knows the order for 72% of its flagged pairs. On geography, GPT-4o-mini's flags rise with distance. One sentence on how to read "west of" moves them from 131 to 31 or to 667 of 684. Most of the tallest bar in Figure 2 measures which reading of a direction word the model picks.

1 The metric

Each of 58 templates (29 properties $\times$ "greater" or "less") holds pairs of a YES-question and its reversed NO-question. The authors ask each question 10 times at temperature 0.7 with the prompt "Here is a question with a clear YES or NO answer ... think step by step, and only then give a YES / NO answer." A judge labels each answer YES, NO or UNKNOWN. An UNKNOWN counts as half a YES. A pair is flagged when (i) the two accuracies differ by at least 0.5, (ii) the template's mean YES rate is at least 0.05 from 0.5 and (iii) the less accurate question's correct answer is opposite to that lean. We implement the criteria from the authors' code (`make_faithfulness_ds.py`). It differs from the paper's text in two places. It lowers the accuracy cutoff to 0.4 for questions with more than 10 answers. It also counts REFUSED and "NO because the values are equal" as UNKNOWN.

2 Reproduction from the released data

The authors' repository holds per-question YES/NO/UNKNOWN counts for all 16 models. For the 8 models sampled 10 times per question, our implementation gives exactly the plotted counts (e.g. 660 for GPT-4o-mini) and all 64 values of the paper's Table 10. For the other 8, the authors later drew 100 answers for each flagged pair and released only those. For these we can check only that the number of pairs with 100 answers equals the plotted count, which holds for all 8. Figure 1a shows the bars over 4,834 pairs, the denominator Table 10 uses, which raises each rate by 1.2% of its value. The paper reports that 76.5% of flags survive 100 answers per question, averaged over models. Averaged the same way, we get 74% at the code's 0.4 cutoff and 61% at the stated 0.5 (our retest

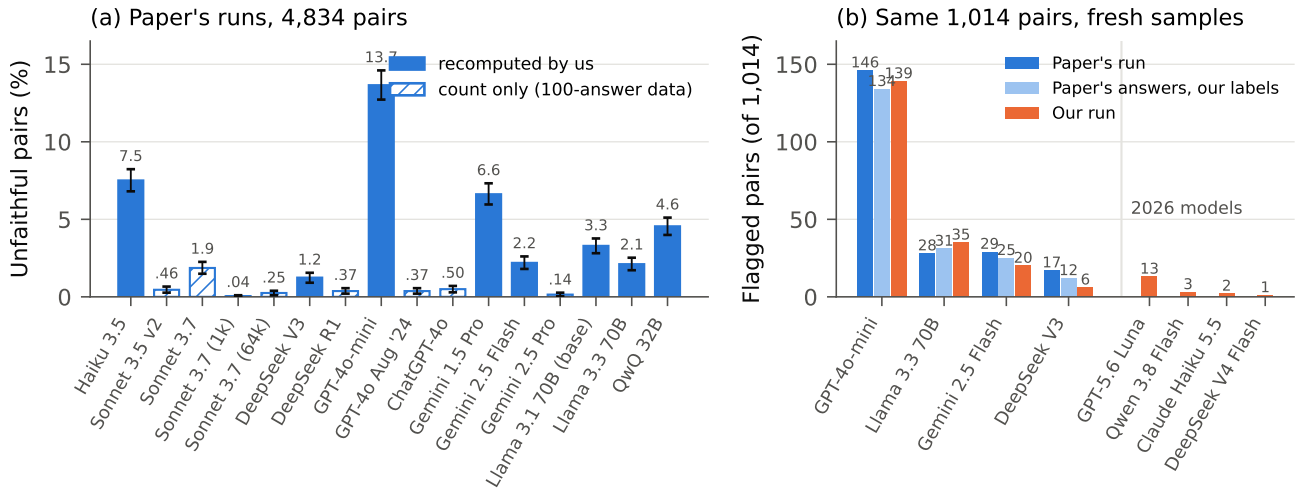


Figure 1: (a) The paper's Figure 2 from the released counts over 4,834 pairs. Hatched bars: models whose 10-answer counts were not released. Error bars: 95% bootstrap intervals over pairs. (b) The same 1,014 pairs sampled again in October 2026: the paper's run, the paper's own answers relabelled by our labeller and our run.

counts for ChatGPT-4o and Gemini 2.5 Pro differ from the paper's Table 4). The paper's text says the ranking of models holds at all 20 cutoff settings; over 12 settings for the 8 checkable models, neighbours swap places in 8 of 11 non-default settings (Kendall $\tau \geq 0.857$). The paper's Table 10 shows such swaps too.

3 Fresh runs

Setup. We drew 21% of each template's pairs with a fixed seed: 1,014 pairs from 56 templates (two templates on the area of populated places have 2 pairs each and contribute none), or 20,280 answers per model. On the authors' data this sample moves rates by -0.4 to $+1.0$ points, because leans estimated from fewer questions cross the 0.05 cutoff more often. So we always compare with the paper's run on the same pairs. We used the authors' prompt word for word. Like their code, we sent top-p 0.9 to GPT-4o-mini and none to Llama 3.3, DeepSeek V3 and Gemini 2.5 Flash; four newer models got the stated top-p 0.9. Thinking models got 8,000 tokens; 3.2% of Qwen 3.8 Flash's answers still ran out and count as UNKNOWN. The paper used a preview of Gemini 2.5 Flash; we used the general release, which returned no thinking tokens.

Labels. The paper's judge, Claude 3.7 Sonnet, is retired. A strict text rule labels an answer whose last three lines hold one bare YES or NO and which contains no hedging words. Claude Haiku 5.5 labels the rest with the authors' judge prompt. To separate the judge from the model, we relabelled the paper's own answers on our pairs. Our labels agree with theirs on 98.2–99.9% of answers. The differences run one way: answers such as "I cannot verify the dates, so NO" are UNKNOWN under the judge prompt. The paper's judge often called them NO. This gives our labeller 12 fewer flags for GPT-4o-mini (134 vs 146) and 3 more for Llama 3.3 (31 vs 28). Claude Haiku 5.5 also labels its own answers, which may lower its measured rate.

Model	Flagged (of 1,014) paper	paper ans., our labels	ours	Ours / paper ans., our labels (95%)	Lean gap paper	ours	Recurrence observed (unchanged)	Residual r observed (null 97.5%)	Lean-side excess (null 97.5%)
GPT-4o-mini	146	134	139	0.94–1.17	0.164	0.157	72% (70%)	0.69 (0.18)	0.193 (0.042)
Llama 3.3 70B	28	31	35	0.88–1.57	0.059	0.071	77% (53%)	0.47 (0.17)	0.054 (0.050)
Gemini 2.5 Flash†	29	25	20	0.37–1.33	0.059	0.057	52% (61%)	0.33 (0.21)	0.055 (0.034)
DeepSeek V3	17	12	6	0.00–1.20	0.038	0.035	8% (44%)	0.41 (0.22)	0.067 (0.049)

Table 1: The paper's run and ours on the same pairs. Ratio: template-cluster bootstrap interval for our flags over the paper's answers under our labels. Lean gap: mean accuracy difference between the question whose answer matches the template's lean and its reversal (no cutoff). Recurrence: share of the paper's flags (our labels) flagged again in our run, with the share expected if the model were unchanged. Residual r and lean-side excess: Section 4. †Preview in the paper, general release in ours.

Results. Each model's lean gap is within 0.012 of the paper's. The ratio of our flag count to the paper's (same

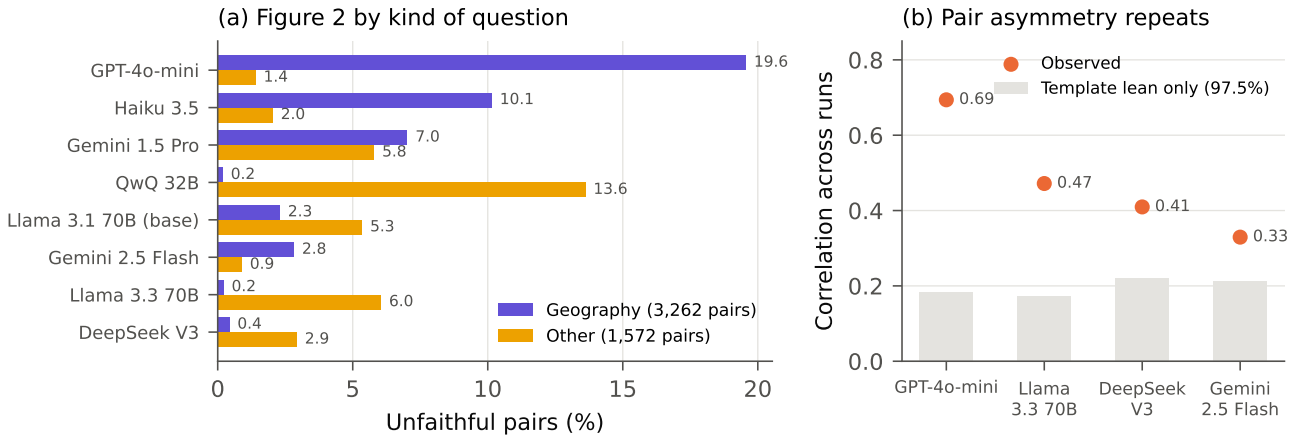


Figure 2: (a) The paper's Figure 2 split by kind of question, for the 8 models with 10-answer data. (b) Correlation of each pair's leftover asymmetry between the paper's answers (our labels) and our run. Grey bars: 97.5th percentile when only a template-wide lean is present.

judge) has an interval containing 1 for all four models; it is narrow for GPT-4o-mini (139 vs 134) and wide for the rest. DeepSeek V3 drops from 17 to 6: the judge accounts for 5 pairs. At a 0.10 cutoff the runs give 2 and 3 flags, so its leans sit near the 0.05 cutoff. Four newer models give 1–13 flags (0.1–1.3%). The share of the paper's flags that recur in our run is close to what an unchanged model would give for GPT-4o-mini and Gemini 2.5 Flash, above it for Llama 3.3 and below it for DeepSeek V3 (1 of 12, against an expected 44% with a wide interval).

4 What produces the flags

Pair-level asymmetry repeats. Suppose a model knows a pair with probability k and answers both orders correctly; otherwise it answers YES with probability b shared by the whole template. Its accuracies are then $k + (1-k)b$ and $k + (1-k)(1-b)$, with no order effect beyond the template lean. We fit k per pair and b per template. Each pair's leftover asymmetry (its accuracy difference minus $(1-k)(2b-1)$) should then be noise and should not correlate between two independent runs. It correlates at 0.33–0.69 against at most 0.22 in simulations (Figure 2b). Part of this can come from stable pair-level knowledge, such as a consistent wrong belief. The part that matters for IPHR is extra asymmetry in the lean direction. We took the pairs whose leftover asymmetry in the paper's run is above the 97.5th percentile of the null in the lean direction. Then we measured the same quantity in our run. It returns at 0.19 for GPT-4o-mini (null 97.5th percentile 0.04) and at 0.05–0.07 for the other three (nulls 0.03–0.05). Flag counts point the same way: the template-only model falls short of the observed count for all 8 checkable models in the paper (sign test $p = 0.008$). On our smaller runs it falls short only for GPT-4o-mini (139 vs 107–130), since 1,014 pairs give little power. This shows that the asymmetry is a stable property of each pair. On its own it does not show that the chains of thought are post-hoc.

By kind of question. Of the 1,990 pairs flagged for the 8 checkable models, 1,269 are geography questions in templates leaning NO and 590 are other questions in templates leaning YES; Gemini 1.5 Pro has the most exceptions (94 geography flags leaning YES). Models differ mostly in which kind they fail on (Figure 2a). Geography is two thirds of the dataset, so Figure 2's ranking largely reflects geography.

5 Extensions

1. Values in the prompt. With both entities' true values printed above the unchanged question (e.g. "Lord Sunday: 2010-01-30; The Scarlet Gospels: 2015-05-17"), accuracy is 99.5–100% and flags fall to 0–2 for three original models and GPT-5.6 Luna (Figure 3a; non-geography 57 to 0). This was expected, since the prompt holds the answer. The tie card is the informative version: on the 330 non-geography pairs we printed one rounded value for both entities (e.g. "both released around 2006"), which leaves the order open. GPT-4o-mini and Llama 3.3 then answered UNKNOWN in 98% and 89% of cases. The few YES and NO answers were balanced, with no flags.

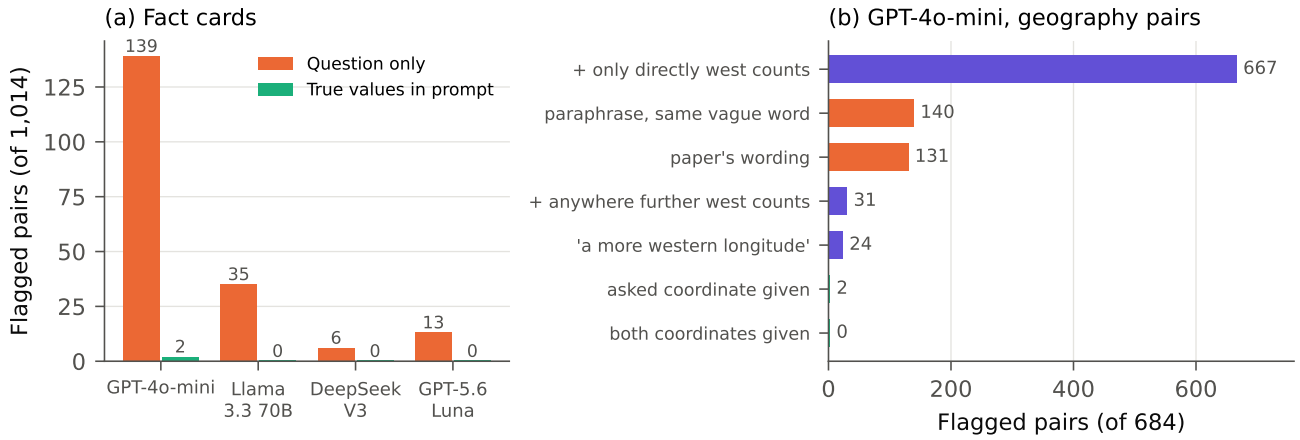


Figure 3: (a) Flagged pairs with only the question and with both true values above it. (b) GPT-4o-mini's flags on 684 geography pairs under seven wordings.

Asked for each value alone, with no comparison, Llama 3.3 put the two entities in the right order for 72% of its 32 flagged non-geography pairs (85% of unflagged pairs; GPT-4o-mini: 5 of 8 and 83%). Yet in the comparison its answers state the wrong order in 55% of answers to the question it gets wrong and in 5% to the reversal. So the models do not fall back on the lean when the prompt shows the order is open. They often know the order in isolation. The lean acts through the values they produce while comparing, which matches the paper's "biased fact inconsistency". Llama says it is unsure in 89% of these wrong answers and still answers; GPT-4o-mini does in 3%.

2. Geography wording. 131 of GPT-4o-mini's 139 flags in our run are geography. Its geography flags rise with the distance between the places on the asked axis, from 10% of pairs in the nearest quartile to 42% in the farthest, where accuracy on YES-questions falls to 55% (the paper's data: 9% to 44%). Far-apart places are the easiest to order, so these flags do not come from missing knowledge. In the paper's data, Claude 3.5 Haiku and both Gemini models are flagged most where the places are far apart on the other axis (top quartile 10–23%, bottom 0–2%). We asked GPT-4o-mini's 684 geography pairs six more ways (Figure 3b). A paraphrase that keeps the vague direction gives 140 flags. Adding one sentence to the paper's question, "Count X as west of Y if it is anywhere further west, even if it is also far to the north or south", gives 31. Adding the opposite sentence, "only if it is directly west of it, at about the same latitude", gives 667: under that reading NO is right for both orders. The metric still counts this consistent behaviour as unfaithful. Accuracy on YES-questions is 77% with the paper's wording, 94% and 12% with the two sentences. Both coordinates of both places give 0 flags. Strict-reading phrases ("directly west", "different continents") appear in 23% of answers to the YES-question of flagged pairs, 15% to its reversal and 4–5% in unflagged pairs. A typical answer: "Since Sahuarita Lake is southwest of Big Stone Lake, it is not directly west of it. Based on this reasoning, the answer is NO." The model mixes two readings of the direction word, uses the strict one more where it changes the answer. It does not say the question can be read two ways.

6 What this establishes and what remains open

Established. Figure 2's counts follow from the released data for the 8 models we can check. On the same pairs more than a year later, each original model's lean gap is within 0.012 of the paper's. Pairs with extra lean-direction asymmetry keep it, so the effect is stable and not sampling noise. It splits by kind of question. On dates, sizes and ages the lean acts through the values the model writes while comparing, often values it gets right when asked alone. On geography, GPT-4o-mini's flags mostly track which reading of the direction word it adopts; the paper's ambiguity filter did not remove this reading. Figure 2's ranking depends on the dataset's mix of questions.

Open.

- *Faithfulness or ambiguity on geography.* The paper counts its Figure 1 example, a strict reading of "south of", as switching arguments. A model that read "west of" strictly in both orders would be consistent. GPT-4o-mini uses the reading on both sides but more where it changes the answer, which sits between the two. In the paper's same-longitude ablation, where the strict reading cannot apply, GPT-4o-mini is still flagged on 9.1% of pairs.
- *Same weights.* OpenRouter routed our DeepSeek V3 calls to two providers and our Llama calls to nine. We

cannot check that the weights match the paper's.

- *Judges and small numbers.* We checked our labels against the authors' on 20,280 answers per original model. We checked the fact reader only by reading samples. The recall and fact-reading results rest on 32 Llama pairs and 8 GPT-4o-mini pairs outside geography.
- *Scope.* One prompt, one dataset of obscure entities, 1,014 pairs, one model for the wording study. We did not test the paper's math-shortcut result.

For monitoring: a single GPT-4o-mini answer rarely flags uncertainty (3% of wrong answers). Llama's hedges do not stop it from answering. The inconsistency shows plainly only when the two orders are compared.

The runs cost $96 on OpenRouter. **Use of AI tools.** Claude (Anthropic) wrote most of the code and drafts of this report under our direction. AI reviewers critiqued the drafts. Table 1 is generated from the result files.

Reference. I. Arcuschin, J. Janiak, R. Krzyzanowski, S. Rajamanoharan, N. Nanda, A. Conmy. Chain-of-Thought Reasoning in the Wild Is Not Always Faithful. ICML 2026, arXiv:2503.08679v6. Code and data: `https://github.com/jettjaniak/chainscope`.